# Operations on Signals:

On the dependent variable:

① Amplitude scaling: $y(t) = c\,x(t)$, $c$ is the scaling factor

$$y[n] = c\,x[n].$$

An electronic amplifier amplifies a signal.

② Addition: Let $x_1(t)$ and $x_2(t)$ be two signals. Then the signal $y(t)$ obtained by the addition of $x_1(t)$ and $x_2(t)$ is defined by

$$y(t) = x_1(t) + x_2(t)$$

A physical example of a device that adds signals is an audio mixer, which combines music and voice signals.

For discrete case: $y[n] = x_1[n] + x_2[n]$

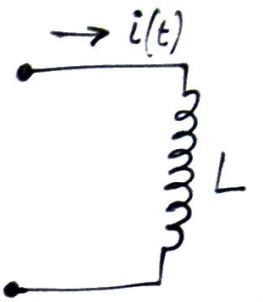
③ Multiplication: Let $x_1(t)$ and $x_2(t)$ denote a pair of continuous-time signals. Then the signal $y(t)$ resulting from multiplication of $x_1(t)$ and $x_2(t)$ is defined by

$$y(t) = x_1(t)\,x_2(t)$$

$$y[n] = x_1[n]\,x_2[n]$$

**Differentiation:** The derivative of a signal $x(t)$ is defined as

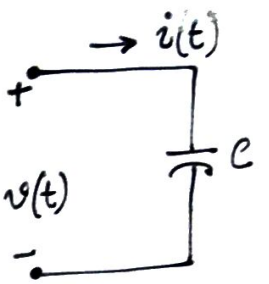
$$y(t) = \frac{d}{dt} x(t)$$




An inductor performs differentiation.

**Integration:** The integration of a signal $x(t)$ is defined as

$$y(t) = \int_{-\infty}^{t} x(\tau)\, d\tau$$




A capacitor performs integration.

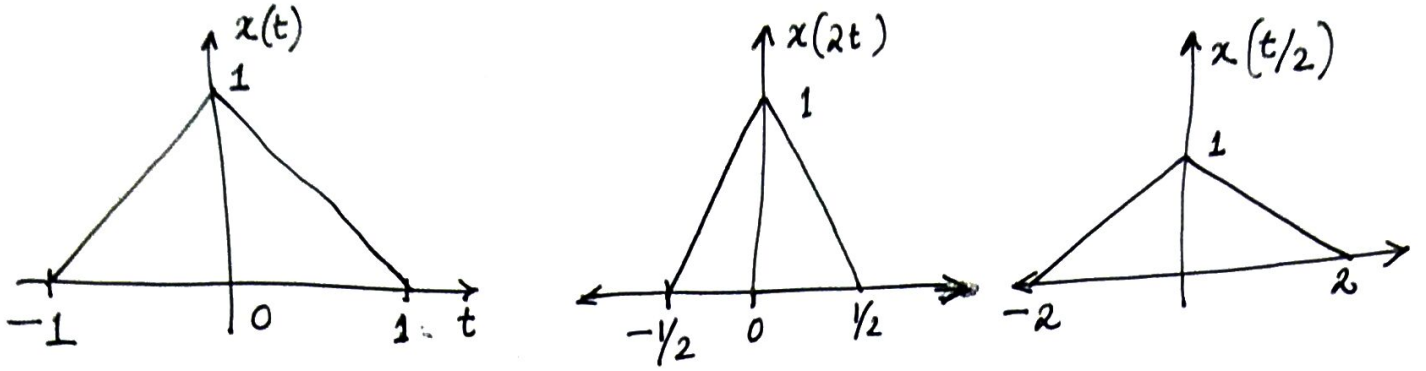
$$v(t) = \frac{1}{C} \int_{-\infty}^{t} i(\tau)\, d\tau$$

**Operations performed on independent variables:**

**Time scaling:** Let $x(t)$ denote a continuous signal. Then the signal $y(t)$ obtained by scaling the independent variable, time $t$, by a factor $a$ is defined by

$$y(t) = x(at)$$

If $a > 1$, the signal $y(t)$ is compressed version of $x(t)$. If $0 < a < 1$, the signal $y(t)$ is an expanded (stretched) version of $x(t)$.




In the discrete case, we write

$$y[n] = x[kn] \ , \ k > 0$$

<u>Reflection:</u> Let $x(t)$ denote a continuous-time signal. Let $y(t)$ denote the signal obtained by replacing time $t$ by $(-t)$; that is

$$y(t) = x(-t)$$ ← Signal $x(t)$ reflected about $t=0$.

Reflected version of even signal is the same signal. Odd signal is the negative of its reflected signal.

<u>Time shifting:</u> Let $x(t)$ denote a continuous-time signal. Then the time-shifted version of $x(t)$ is defined by

$$y(t) = x(t - t_0) \ , \ t_0 \in \mathbb{R}^+$$

If $t_0 > 0$, signal shifts to right.

If $t_0 < 0$, signal shifts to left.

$$y[n] = x[n-m] \ , \ n, m \in \mathbb{Z}$$

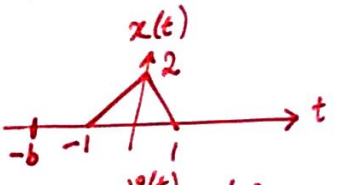
# Precedence rule for time shifting and time scaling:

$$y(t) = x(at-b)$$

$$\therefore \boxed{y(0) = x(-b)} \quad \& \quad \boxed{y(b/a) = x(0)}$$

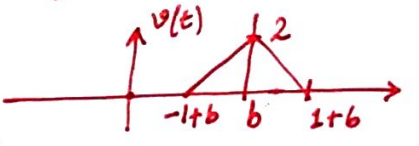
Used for checking

* Time shifting changes $t$ by $t-b$
* Time scaling changes $t$ by $at$.

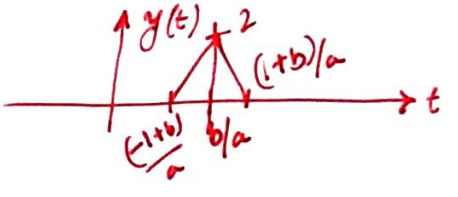



## First method:

① Time shift: $v(t) = x(t-b)$



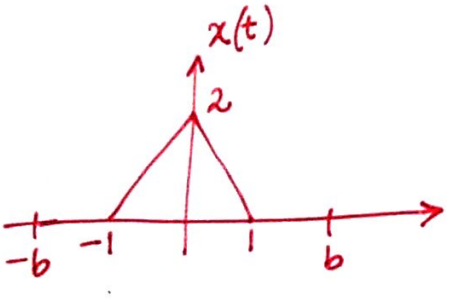

② Time scale: $y(t) = v(at) = x(at-b)$

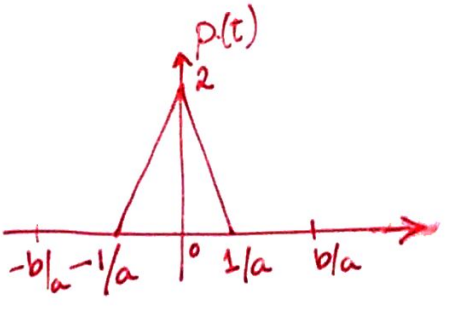


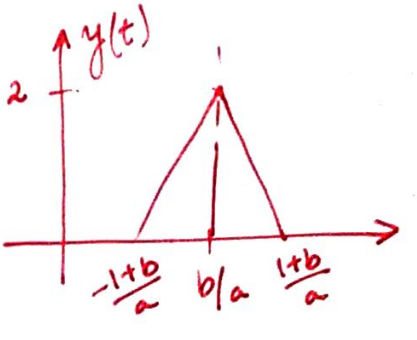

## 2nd method:

① Time scale: $p(t) = x(at)$

② Time shift: $y(t) = p(t-b/a) = x(a(t-b/a)) = x(at-b)$



# Elementary Signal:

## ① Exponential Signals:

A real exponential signal is written as

$$x(t) = B e^{at}$$

where B and a are real parameters.

$x[n] = B r^n$
$0 < r < 1$ fall
$r > 1$ fall
$r < 0$, n even +ve
n odd −ve

$a < 0 \longrightarrow$ Decaying exponential

$a > 0 \longrightarrow$ Growing exponential

e.g. Decay rate of radioactive material:

* Relationship between quantity and time of radioactive decay —

(Chemical + Physics)

$$N(t) = e^{-\lambda t} N_0$$

$\lambda$: decay constant

$N(t)$: number of undecayed nuclei at time t

$N_0$: number of undecayed nuclei at time $t=0$

* Exponential convergence of algorithms:

The error converges exponential

* Crack-growth rate (Paris-Erdogan equation)

$$\frac{da}{dN} = C(\Delta K)^m$$

$\Delta K$: stress intensity factor range of a load cycle ($\Delta K$)

a → crack length
da/dN → fatigue crack growth for single load cycle N.

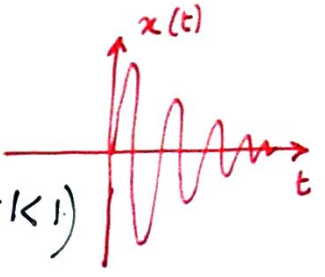
② Sinusoidal Signals:

$$x(t) = A\cos(\omega t + \phi)$$

③ Exponentially Damped Sinusoidal signals:

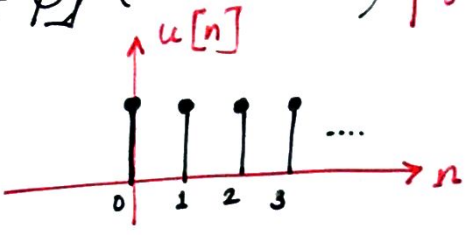
$$x(t) = Ae^{-at}\sin(\omega t + \phi) \qquad a > 0$$

$$x[n] = Br^n \sin[\Omega n + \phi] \quad (\text{Decay}, 0 < |r| < 1)$$

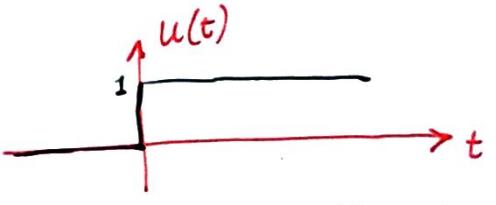



④ Step functions:

$$u[n] = \begin{cases} 1, & n \geq 0 \\ 0, & n < 0 \end{cases}$$



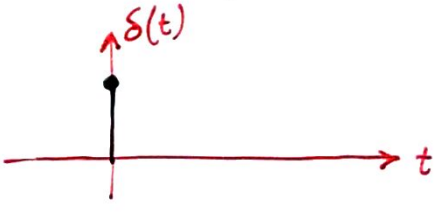

$$u(t) = \begin{cases} 1, & t > 0 \\ 0, & t < 0 \end{cases}$$




e.g. a battery source, (Important, because it reveals the response of a system to an abrupt change in signal.)

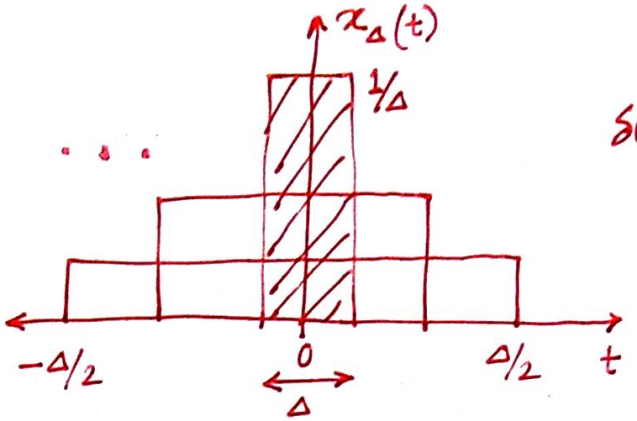
⑤ Impulse functions:

Discrete: $\delta[n] = \begin{cases} 1, & n = 0 \\ 0, & n \neq 0 \end{cases}$




Continuous: $\left.\begin{array}{l} \delta(t) = 0 \quad \text{for } t \neq 0 \\ \text{and} \displaystyle\int_{-\infty}^{\infty} \delta(t)\,dt = 1 \end{array}\right\}$ Dirac delta function

Visualization of discreate impulse is easy but for continuous case it requires some work.




$\delta(t) = \lim_{\Delta \to 0} x_\Delta(t)$,

where $x_\Delta(t)$ is any pulse that is an even function of $t$ with duration $\Delta$ and unit area.

Relation between $u(t)$ & $\delta(t)$:

$$\boxed{\delta(t) = \frac{d}{dt} u(t)} \qquad \boxed{u(t) = \int_{-\infty}^{t} \delta(t)\,dt}$$

## ⑥ Ramp function:

$$r(t) = \begin{cases} t, & t \geq 0 \\ 0, & t < 0 \end{cases} \qquad \text{'OR'} \qquad r(t) = t\,u(t)$$

If $f(t)$ represents angular displacement of a shaft, then the constant-speed rotation of the shaft provides a representation of the ramp function.

As a test signal, the ramp function enables us to evaluate how a continuous-time system would respond to a signal that increases linearly with time.

$$r[n] = \begin{cases} n, & n \geq 0 \\ 0, & n < 0 \end{cases}$$